

NAME	KEN KIBUGI MUIRURI
TRACKING NUMBER	CS-SA12-26014
COHORT	CYBERSHUJAA COHORT 1-2026
TASK	WINDOWS FUNDAMENTALS REPORT

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WINDOWS FUNDAMENTALS REPORT

Introduction

The Windows Fundamentals module provided a structured introduction to the Windows operating system from both a functional and security perspective. The course focused on understanding Windows architecture, system management tools, and security concepts that are essential for cybersecurity analysis.

Here is a shareable link to my completed module:

<https://academy.hackthebox.com/achievement/2380391/49>

1.1: Introduction to windows

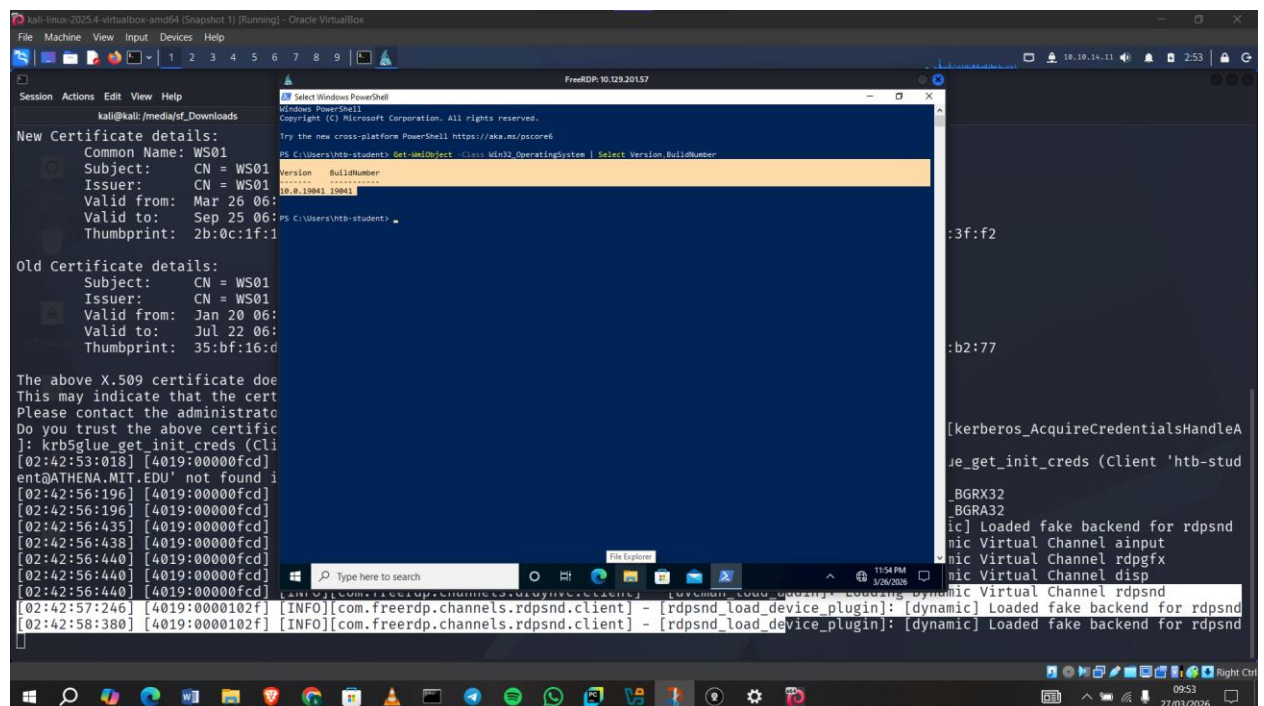
I learned the role of Windows as a widely used operating system in both personal and enterprise environments. Understanding Windows is critical in cybersecurity because it is a common target for attacks and the primary OS used in many organizations.

Question1: What is the Build Number of the target workstation?

Answer: I ran **Get-WmiObject -Class win32_OperatingSystem | select Version,BuildNumber** and found **19041**

Question2: Which Windows NT version is installed on the workstation?

Answer: **Windows 10**

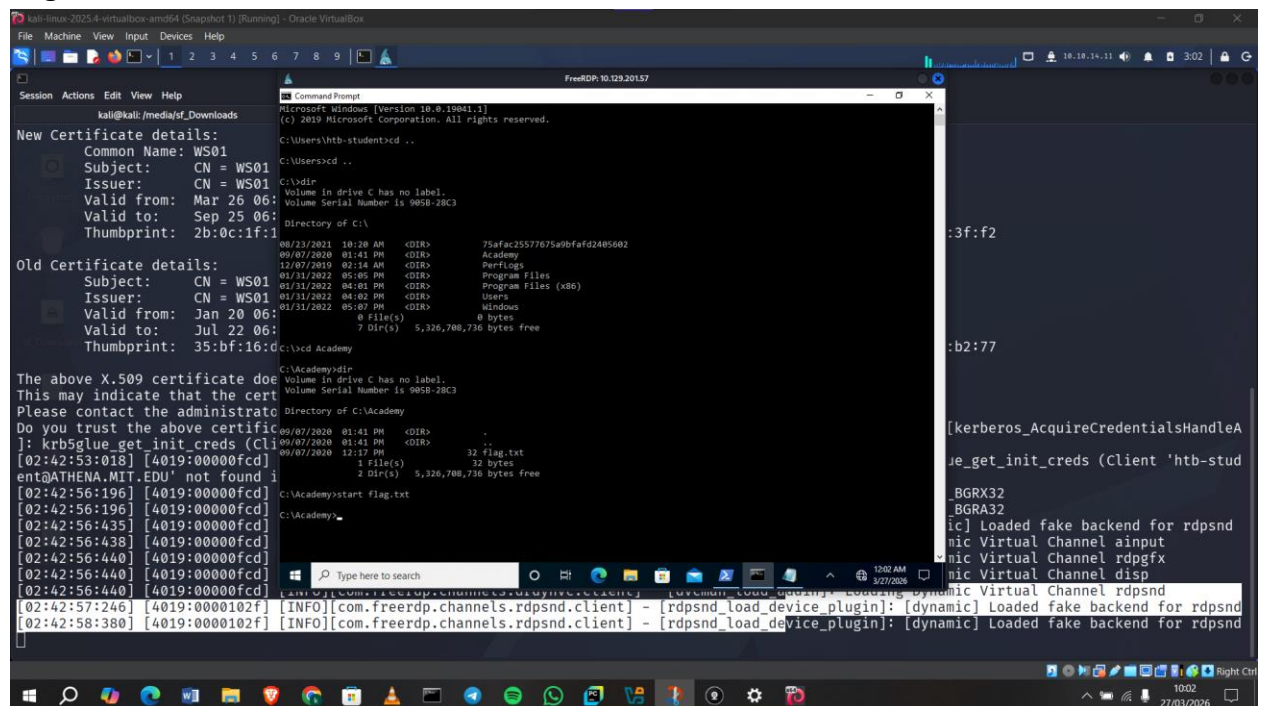


1.2: Operating System structure

This module introduced the Windows root directory and the structure of the boot partition. I learned the purpose of key system and user directories such as Program Files, Users, Windows, System32, AppData, and WinSxS. I also practiced navigating the file system using the dir command.

Question: Find the non-standard directory in the C drive. Submit the contents of the flag file saved in this directory.

Answer: I moved to C, located the file as **Academy** and opened the flag file using **start flag.txt**.

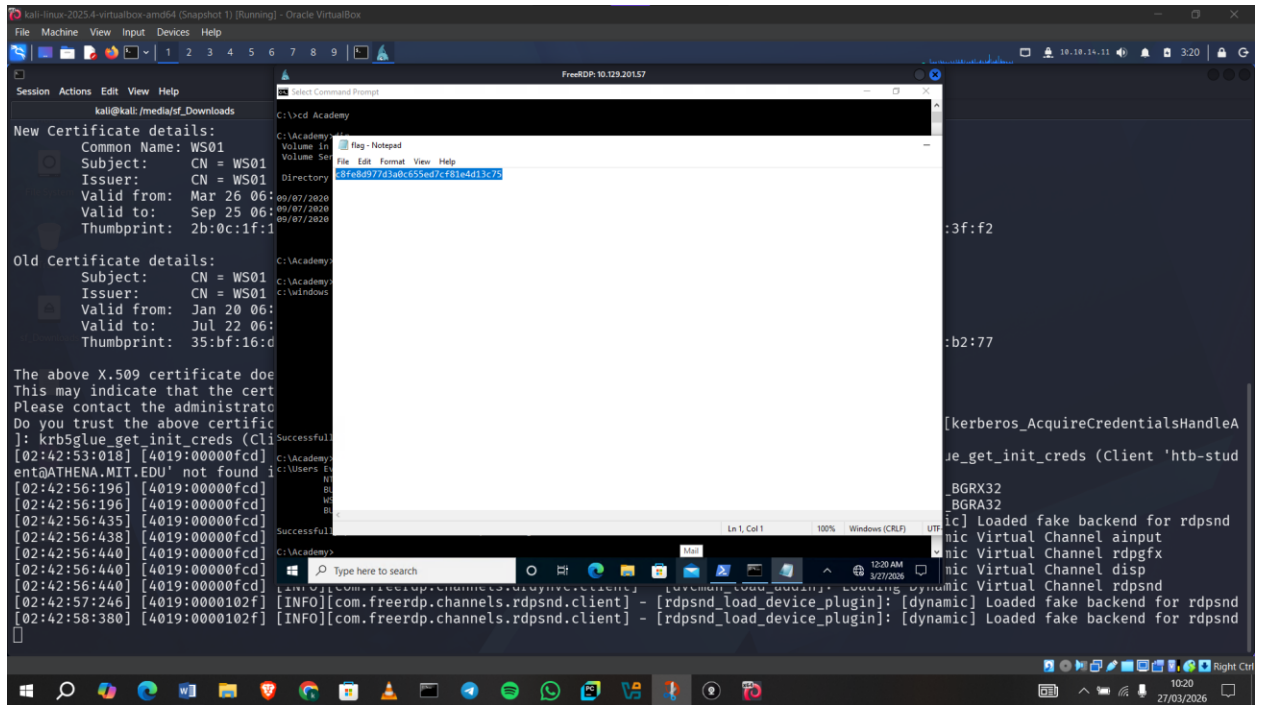


```
kali@kali:~/media/xf_Downloads$ cd ..
New Certificate details:
  Common Name: WS01
  Subject: CN = WS01
  Issuer: CN = WS01
  Valid from: Mar 26 06:
  Valid to: Sep 25 06:
  Thumbprint: 2b:0c:1f:1

Old Certificate details:
  Subject: CN = WS01
  Issuer: CN = WS01
  Valid from: Jan 20 06:
  Valid to: Jul 22 06:
  Thumbprint: 35:bf:16:0

The above X.509 certificate does not appear to be a valid certificate.
This may indicate that the certificate is not yet issued or has expired.
Please contact the administrator for more information.
Do you trust the above certificate? [Y/n]
]: krb5glue_get_init_creds (Client 'htb-student@ATHENA.MIT.EDU' not found)
[02:42:53:018] [4019:00000fcd] [INFO] com.freerdp.channels.rdpsnd.client - [rdpsnd_load_device_plugin]: [dynamic] Loaded fake backend for rdpsnd
[02:42:56:196] [4019:00000fcd] [INFO] com.freerdp.channels.rdpsnd.client - [rdpsnd_load_device_plugin]: [dynamic] Loaded fake backend for rdpsnd
[02:42:56:196] [4019:00000fcd] [INFO] com.freerdp.channels.rdpsnd.client - [rdpsnd_load_device_plugin]: [dynamic] Loaded fake backend for rdpsnd
[02:42:56:435] [4019:00000fcd] [INFO] com.freerdp.channels.rdpsnd.client - [rdpsnd_load_device_plugin]: [dynamic] Loaded fake backend for rdpsnd
[02:42:56:438] [4019:00000fcd] [INFO] com.freerdp.channels.rdpsnd.client - [rdpsnd_load_device_plugin]: [dynamic] Loaded fake backend for rdpsnd
[02:42:56:440] [4019:00000fcd] [INFO] com.freerdp.channels.rdpsnd.client - [rdpsnd_load_device_plugin]: [dynamic] Loaded fake backend for rdpsnd
[02:42:56:440] [4019:00000fcd] [INFO] com.freerdp.channels.rdpsnd.client - [rdpsnd_load_device_plugin]: [dynamic] Loaded fake backend for rdpsnd
[02:42:57:246] [4019:0000102f] [INFO] com.freerdp.channels.rdpsnd.client - [rdpsnd_load_device_plugin]: [dynamic] Loaded fake backend for rdpsnd
[02:42:58:380] [4019:0000102f] [INFO] com.freerdp.channels.rdpsnd.client - [rdpsnd_load_device_plugin]: [dynamic] Loaded fake backend for rdpsnd
```

The flag was: **c8fe8d977d3a0c655ed7cf81e4d13c75**

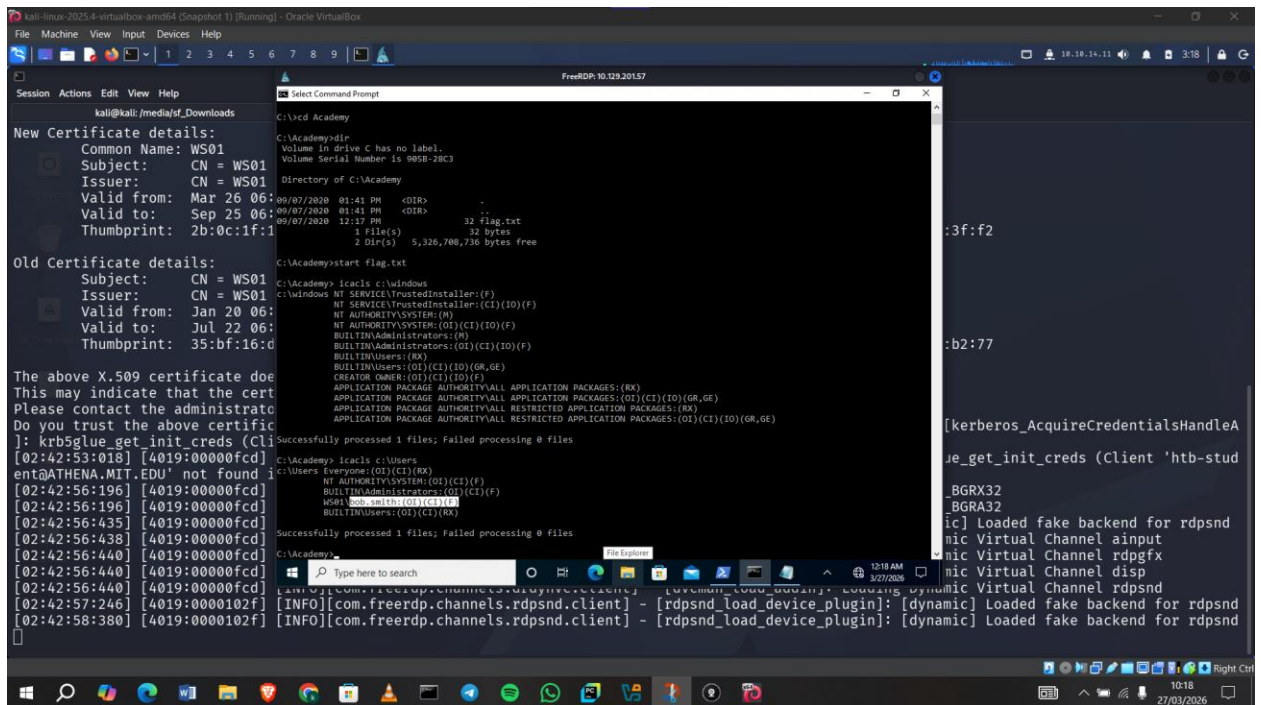


1.3: File system

The module introduced the main Windows file systems, focusing on **NTFS**, which is the default for modern Windows. Understanding these file systems and permissions is essential for system security, access control, and managing files in Windows environments.

Question: What system has full control over the c:\users directory?

Answer: I ran **icacls c:\Users** and found the user: **bob.smith**



1.4: NTFS vs Share Permissions

I learned the difference between NTFS permissions and share permissions, and how effective permissions are determined when both are applied. This is important in understanding access control, privilege abuse, and misconfigurations that can lead to security vulnerabilities.

Question1: What protocol discussed in this section is used to share resources on the network using Windows? (Format: case sensitive)

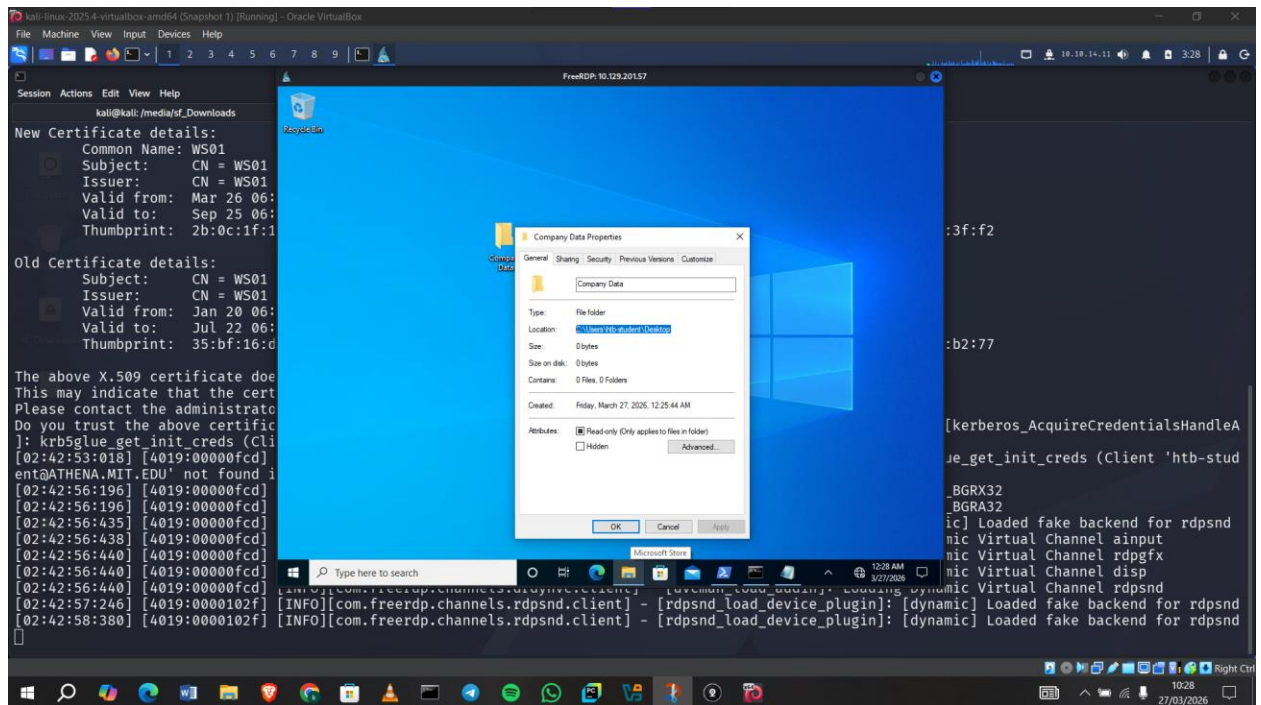
Answer: **SMB**

Question2: What is the name of the utility that can be used to view logs made by a Windows system?

Answer: **Event Viewer**

Question3: What is the full directory path to the Company Data share we created?

Answer: **C:\Users\htb-student\Desktop\Company Data**



1.5: Windows Services & Processes

This module introduced Windows services and processes, how they start, and how they run in the background. I learned how services differ from processes and how they are managed by the system.

Question: Identify one of the non-standard update services running on the host. Submit the full name of the service executable.

Answer: I opened **Task manager>Details** and found it as: **foxitreaderupdateservice.exe**

1.8: Interacting with the Windows Operating System

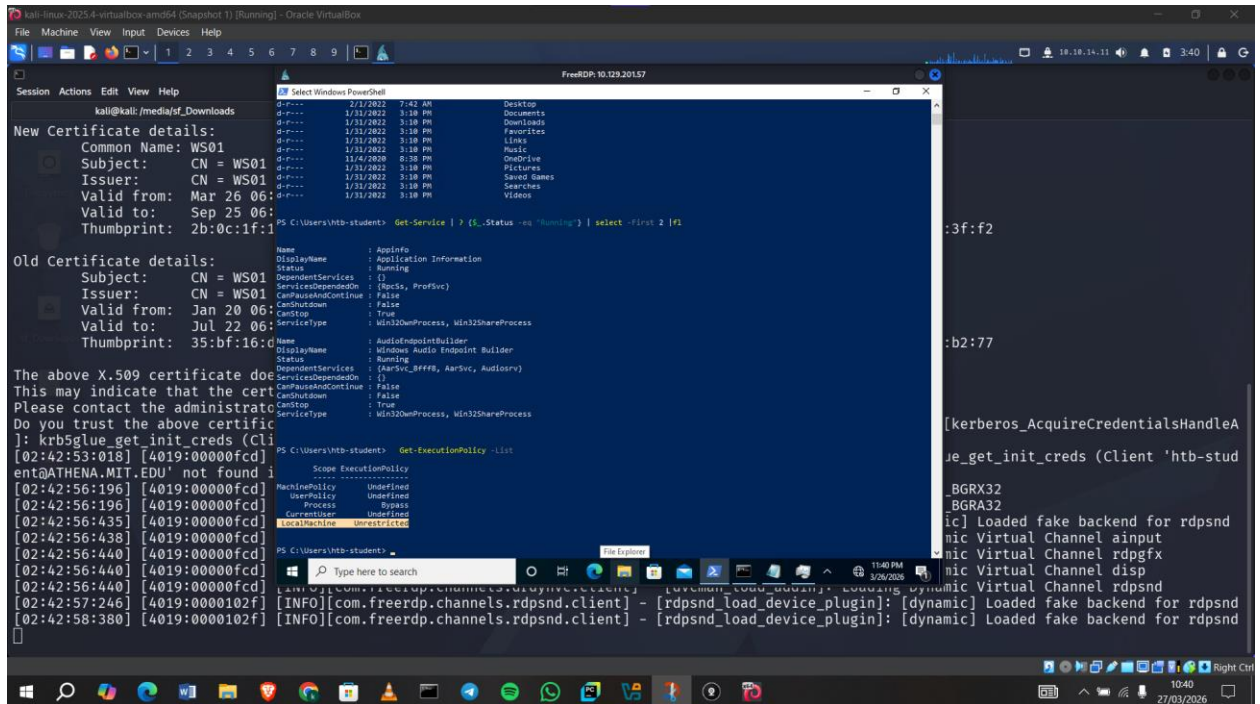
I practiced interacting with Windows using different tools, including the GUI, Command Prompt, and PowerShell.

Question1: What is the alias set for the ipconfig.exe command?

Answer: **ifconfig**

Question2: Find the Execution Policy set for the LocalMachine scope.

Answer: I ran **Get-ExecutionPolicy -List** and found the policy as **Unrestricted**.

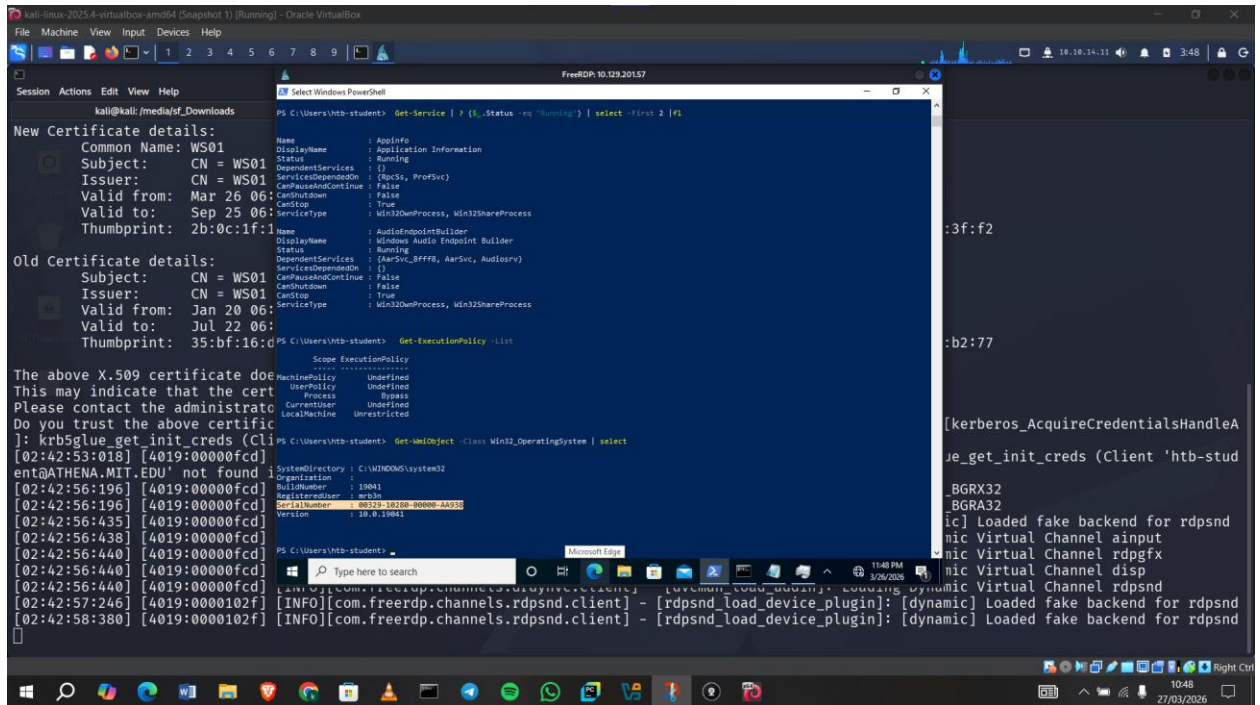


1.9: Windows Management Instrumentation (WMI)

I learned how WMI allows administrators and analysts to query system information, manage resources, and automate administrative tasks. WMI is a powerful tool that can be used for both system management and security monitoring.

Question: Use WMI to find the serial number of the system.

Answer: I ran **Get-WmiObject -Class Win32_OperatingSystem | select** and found:
00329-10280-00000-AA938



The screenshot shows a Kali Linux terminal window with the following content:

```
kali@kali:~/media/sf_downloads$ PS C:\Users\htb-student> Get-Service | ? {$_.Status -eq 'Running'} | select -first 2 | fl
Name           : AppInfo
DisplayName     : Application Information
Status         : Running
DependentServices : {}
ServiceName     : (Ncs, ProfSvc)
CanPauseAndContinue : False
CanShutdown    : False
CanStop        : True
ServiceType     : Win32OwnProcess, Win32ShareProcess

Name           : AudioEndpointBuilder
DisplayName     : Windows Audio Endpoint Builder
Status         : Running
DependentServices : {}
ServiceName     : (AarSvc, ProfSvc, AudioSvc)
CanPauseAndContinue : False
CanShutdown    : False
CanStop        : True
ServiceType     : Win32OwnProcess, Win32ShareProcess

PS C:\Users\htb-student> Get-ExecutionPolicy -List
Scope ExecutionPolicy
-----
MachinePolicy Undefined
UserPolicy    Undefined
Process       Bypass
CurrentUser   Undefined
LocalMachine  Unrestricted

PS C:\Users\htb-student> Get-WmiObject -Class Win32_OperatingSystem | select
SystemDirectory : C:\WINDOWS\system32
Organization     :
BuildNumber      : 19041
RegisteredOwner  :
SerialNumber     : 00329-10280-00000-AA938
Version          : 10.0.19041

[02:42:53:018] [4019:00000fcd] ent@ATHENA.MIT.EDU' not found
[02:42:56:196] [4019:00000fcd]
[02:42:56:196] [4019:00000fcd]
[02:42:56:435] [4019:00000fcd]
[02:42:56:438] [4019:00000fcd]
[02:42:56:440] [4019:00000fcd]
[02:42:56:440] [4019:00000fcd]
[02:42:56:440] [4019:00000fcd]
[02:42:57:246] [4019:0000102f] [INFO][com.freerdp.channels.rdpnd.client] - [rdpsnd_load_device_plugin]: [dynamic] Loaded fake backend for rdpnd
[02:42:58:380] [4019:0000102f] [INFO][com.freerdp.channels.rdpnd.client] - [rdpsnd_load_device_plugin]: [dynamic] Loaded fake backend for rdpnd
```

1.10: Microsoft Management Console

This module introduced MMC and its snap-ins, which are used to manage Windows components such as services, users, and event logs. MMC provides a centralized way to manage and monitor Windows systems.

1.11: Windows Subsystem for Linux

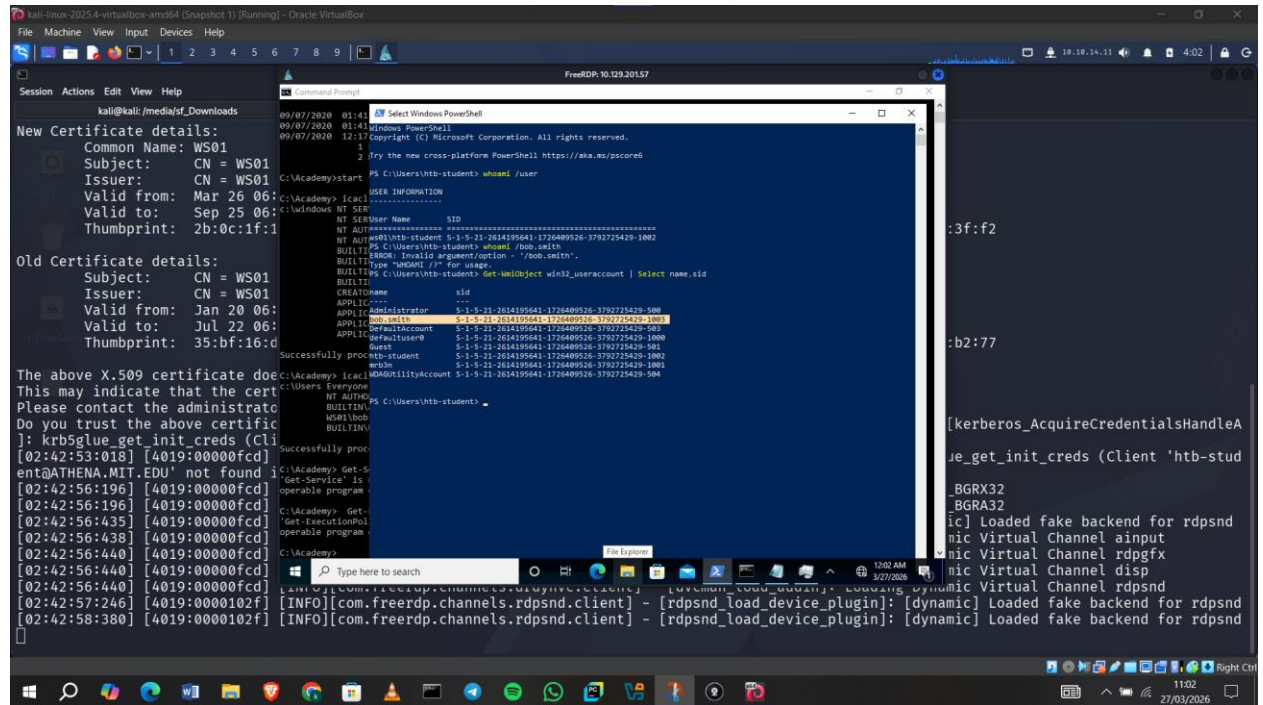
I learned about WSL and how it allows Linux environments to run directly on Windows. This is useful for developers and security professionals who require Linux tools while working within a Windows environment.

1.12: Windows Security

This section focused on Windows security mechanisms, including authentication, permissions, services, and security controls. I learned how Windows implements security at multiple levels to protect systems and data.

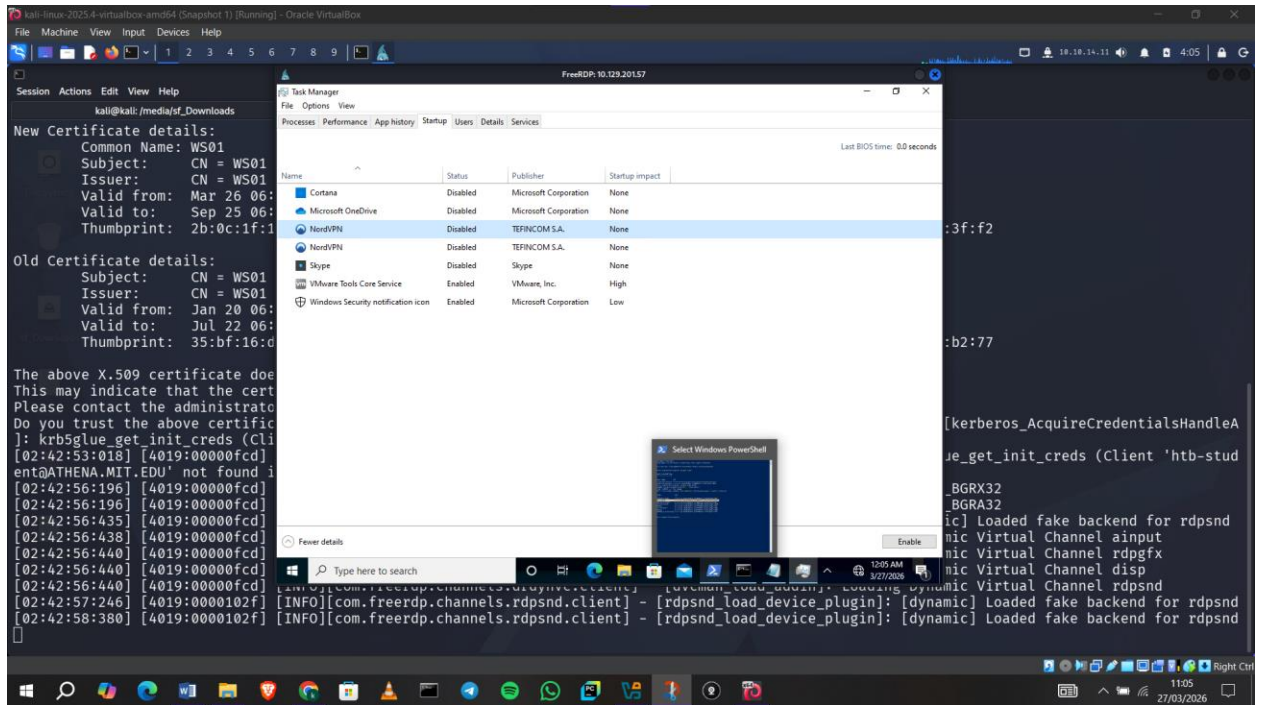
Question1: Find the SID of the bob.smith user.

Answer: I ran **Get-WmiObject win32_useraccount | Select name,sid** and found: **S-1-5-21-2614195641-1726409526-3792725429-1003**



Question2: What 3rd party security application is disabled at startup for the current user?

Answer: I opened **Task Manager>Startup** and found: **NordVPN**



1.13: Skills Assessment - Windows Fundamentals

Question1: What is the name of the group that is present in the Company Data Share Permissions ACL by default?

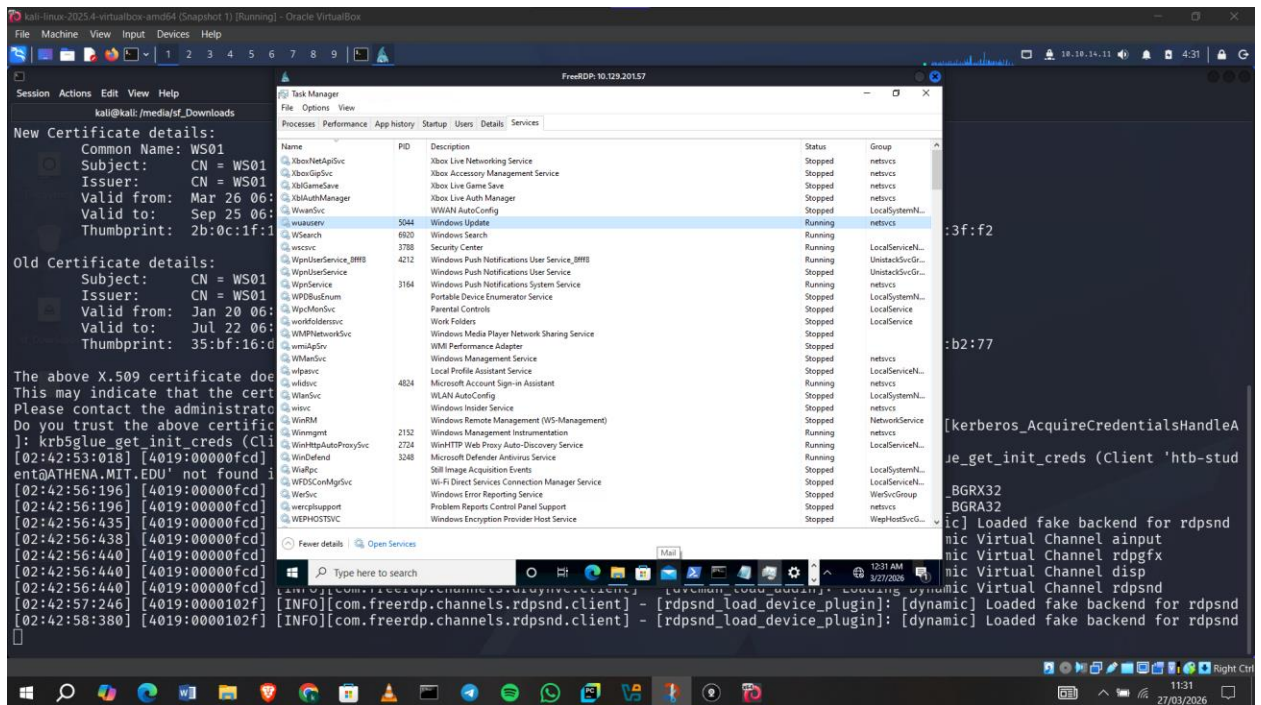
Answer: **everyone**

Question2: What is the name of the tab that allows you to configure NTFS permissions?

Answer: **security**

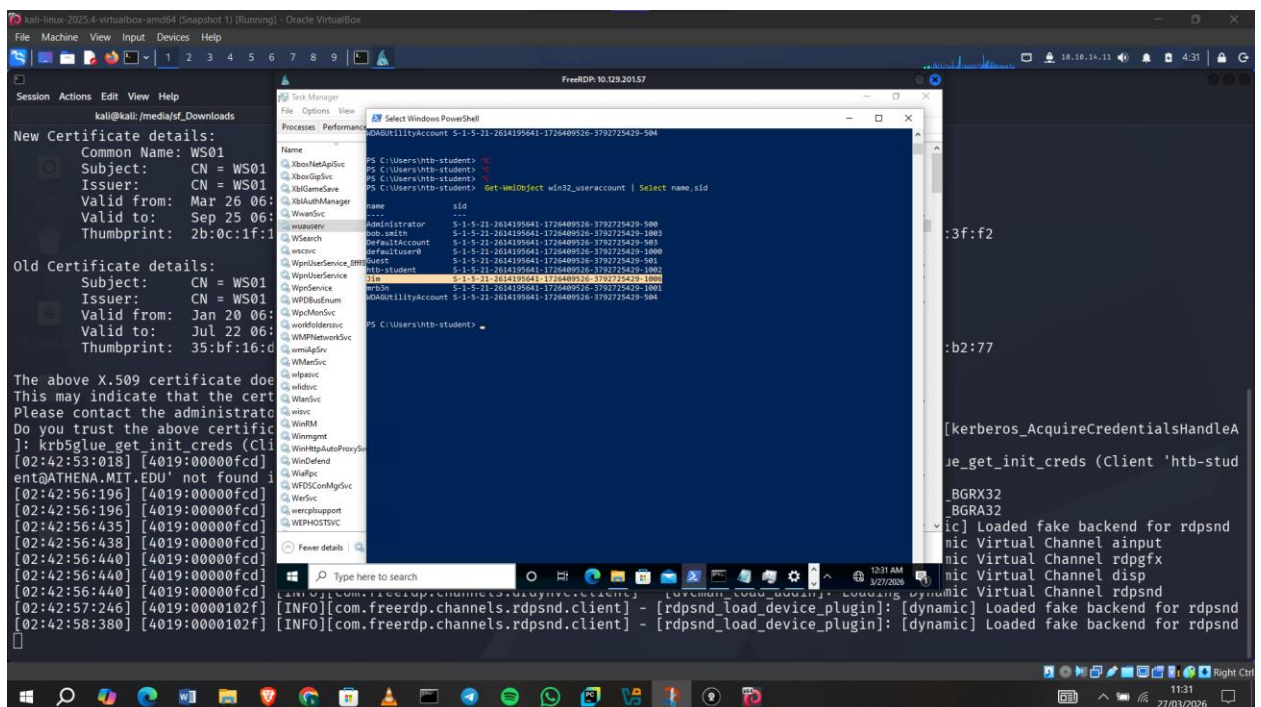
Question3: What is the name of the service associated with Windows Update?

Answer: I opened **Task Manager> Services** and found: **wuauserv**



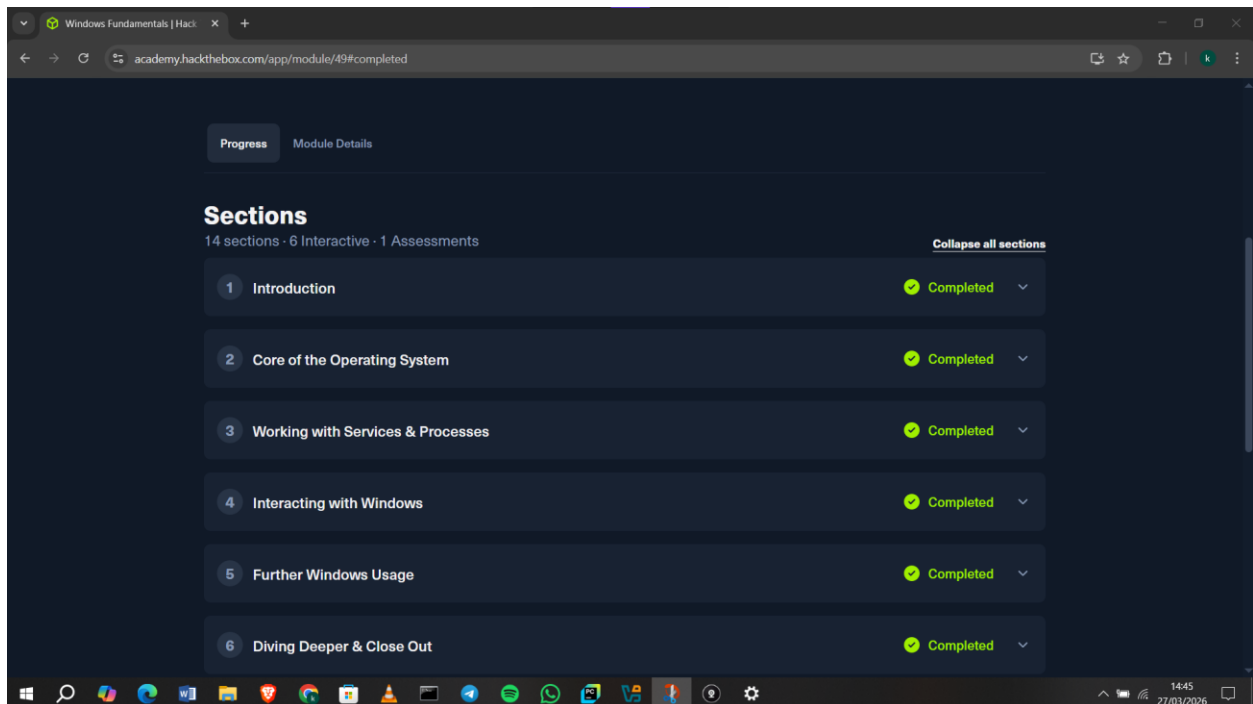
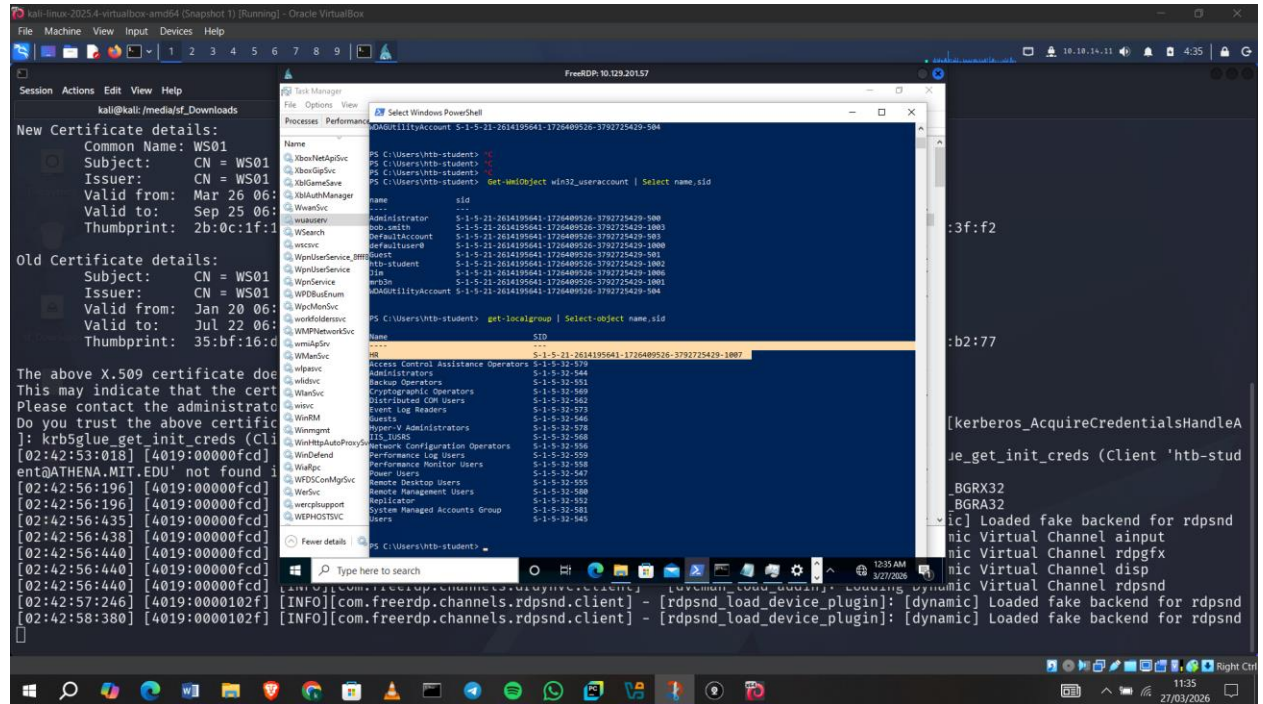
Question4: List the SID associated with the user account Jim you created.

Answer: I ran **Get-WmiObject win32_useraccount** | **Select name,sid** and found: **S-1-5-21-2614195641-1726409526-3792725429-1006**



Question5: List the SID associated with the HR security group you created.

Answer: I ran **get-localgroup | Select-object name,sid** and found: **S-1-5-21-2614195641-1726409526-3792725429-1007**



Conclusion

Completing the Windows Fundamentals module gave me a strong understanding of how Windows operates internally and how it is analyzed from a cybersecurity perspective. The skills gained form a solid foundation for more advanced topics in my cybersecurity journey.